DNS in Windows Server

Task 1: Open Server Manager on your Windows Server VM, add the DNS Server role, and take a screenshot showing the DNS role installed.

The DNS Server role was added through Server Manager (Manage > Add Roles and Features > Role-based installation > DNS Server). The screenshot shows DNS already installed, in the Server. Server Manager navigation pane and SERVER01 online under the DNS servers list.

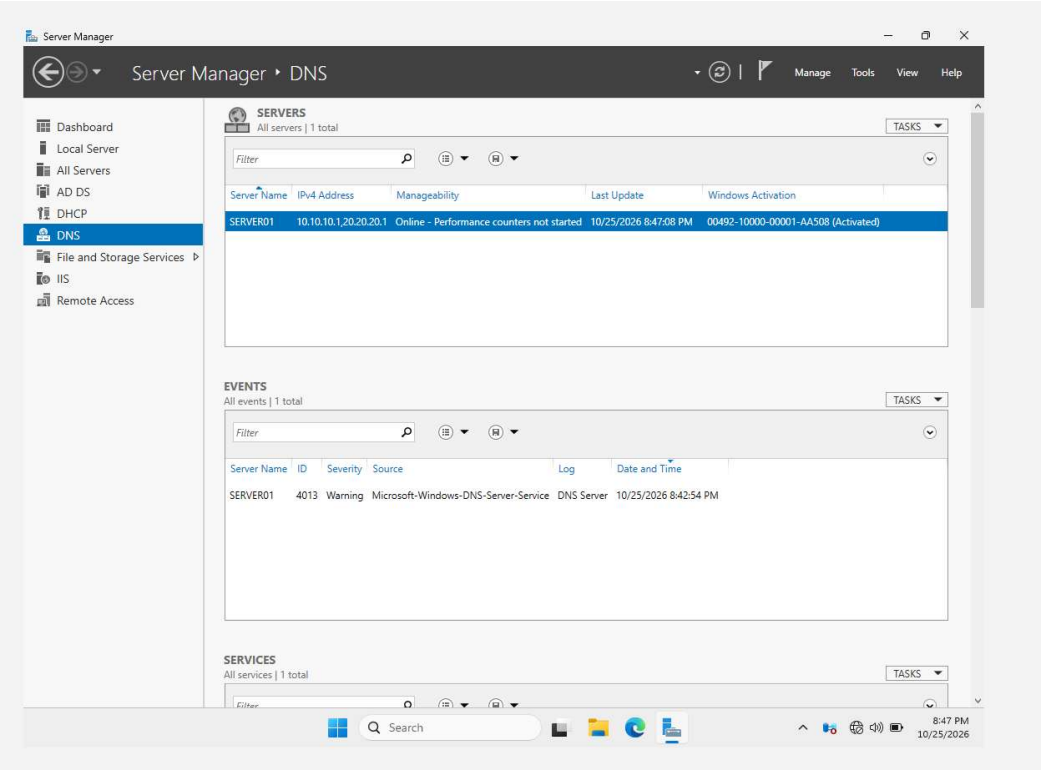


Figure 2: Server Manager showing the DNS role installed

Task 2: In DNS Manager, create a new Forward Lookup Zone named foodies.local and add an A record for zomato.foodies.local pointing to 192.168.1.50.

1. Open DNS Manager.
2. Right-click Forward Lookup Zones and choose New Zone.
3. I. Selected Primary zone and enter the zone name foodies.local.
4. I, Right-click the foodies.local zone and choose New Host (A or AAAA).
5. Enter the name **zomato** and the IP address **192.168.1.50**, then click Add Host.

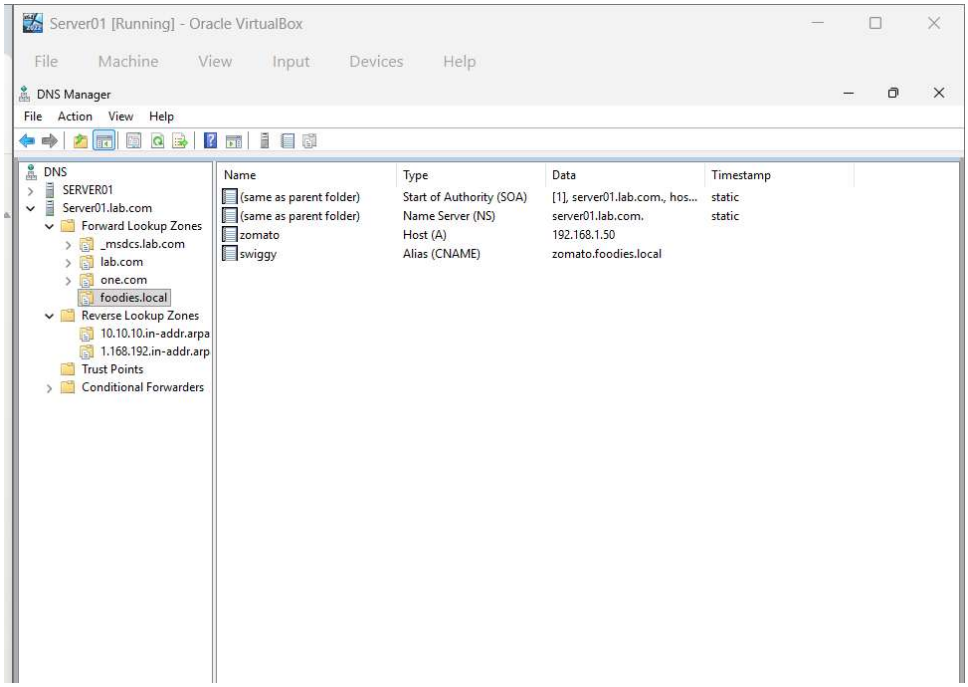


Figure 3: foodies.local zone with the zomato A record and the swiggy CNAME

Task 3: Add a CNAME record in the foodies.local zone so that swiggy.foodies.local points to zomato.foodies.local. Hint: Use the 'New Alias (CNAME)' option and test both names with nslookup.

1. I had Right-click the foodies.local zone and choose New Alias (CNAME).
2. Entered the alias name **swiggy**.
3. Entered the target host as **zomato.foodies.local**

Figure 3 above shows the resulting swiggy alias (CNAME) pointing to zomato.foodies.local.

Task 4: Simulate a DNS resolution: On a client machine, use nslookup to resolve swiggy.foodies.local and zomato.foodies.local, then explain what happens and why in 3-4 lines.

On Client01, nslookup was used to resolve both names against the DNS server at 10.10.10.1.

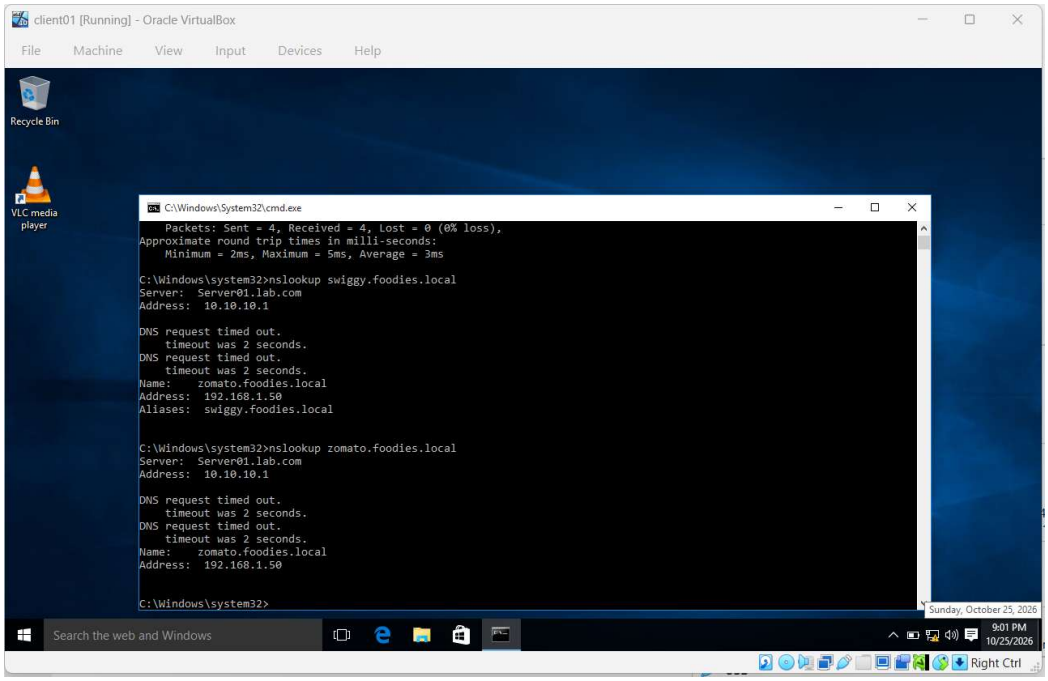


Figure 4: nslookup results on Client01

Query	Result
swiggy.foodies.local	Name: zomato.foodies.local, Address: 192.168.1.50, Aliases: swiggy.foodies.local
zomato.foodies.local	Name: zomato.foodies.local, Address: 192.168.1.50

Explanation

When I look up zomato.foodies.local, DNS gives me its IP address, 192.168.1.50, straight away. When I look up swiggy.foodies.local, DNS sees that swiggy is just another name for zomato. It goes to zomato and gives me the same IP address. So both names lead to the same address. If zomato's IP ever changes, I only have to fix one record and swiggy will follow it.

Observation

The first time I ran nslookup, it showed a few "DNS request timed out" messages before the answer came. The final answers were still correct. This probably happened because the server has two network adapters, so the reply may have come back from a different address than the one the client asked.

Additional Verification:

I also created a reverse lookup zone (1.168.192.in-addr.arpa) with a PTR record. This works the opposite way to normal DNS: if I look up the IP address 192.168.1.50, DNS tells me the name zomato.foodies.local.

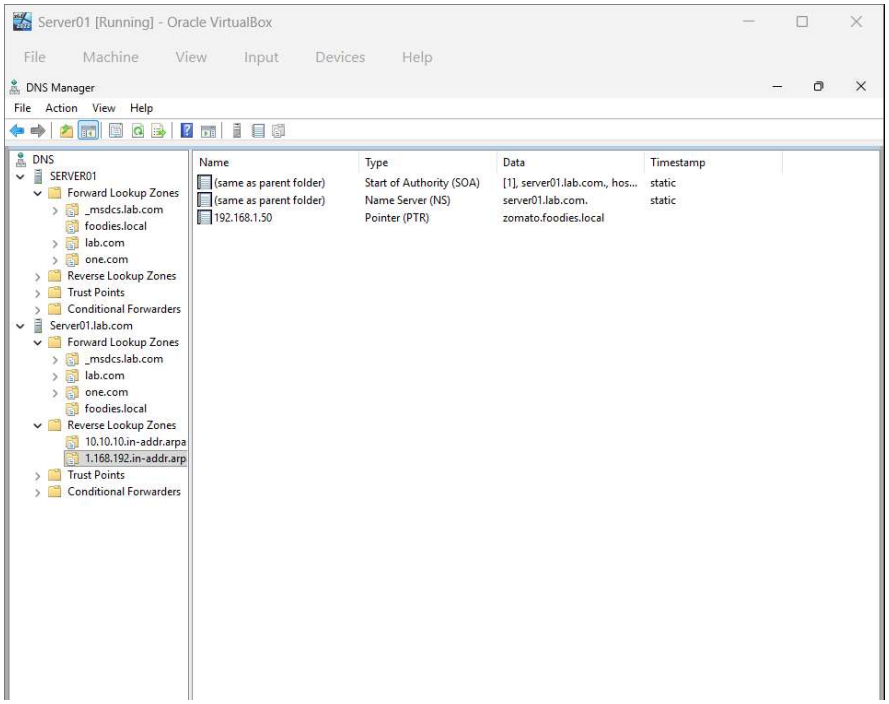


Figure 5: Reverse lookup zone with the PTR record for 192.168.1.50

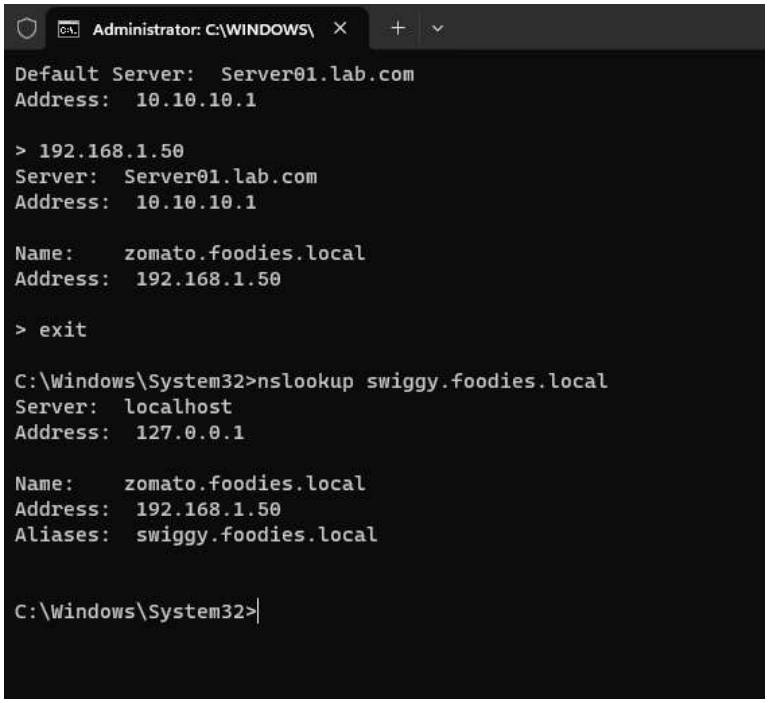


Figure 6: Reverse lookup of 192.168.1.50 returning zomato.foodies.local